**ACADEMIC CITY UNIVERSITY**

**END OF FIRST SEMESTER EXAMINATION – 2025/2026**
**D5203: STATISTICAL METHODS FOR DATA SCIENCE**

TOPIC :

# FRAUDX AI: AN EXPLAINABLE AND ADAPTIVE MACHINE LEARNING FRAMEWORK FOR CREDIT CARD FRAUD DETECTION

**AYEBIE GRAM MESCHAC**

**200250045**

# Table of Contents

ABSTRACT ---------- 3

ACRONYMS ---------- 3

I. INTRODUCTION ---------- 5

II. PROBLEM STATEMENT: ESTABLISHING THE RESEARCH GAP ---------- 7

III. OBJECTIVES ---------- 9

IV. SIGNIFICANCE OF STUDY ---------- 10

V. LITERATURE REVIEWS ---------- 12

VI. METHODOLOGY ---------- 17

VII . RESULTS AND DISCUSSIONS ---------- 20

VIII. DISCUSSION , SUMMARY , CONCLUSION AND RECOMMENDATION ---------- 30

IX. REFERENCES AND APPENDIX ---------- 31

# ACRONYMS

XGBoost : Extreme Gradient Boosting (Algorithme de boosting performant).

XAI : Explainable Artificial Intelligence (Intelligence Artificielle Explicable).

SHAP : SHapley Additive exPlanations (Méthode d'explication basée sur la théorie des jeux).

LIME : Local Interpretable Model-agnostic Explanations (Explication locale des modèles).

SMOTE : Synthetic Minority Over-sampling Technique (Sur-échantillonnage des données minoritaires).

SMOTEENN : SMOTE with Edited Nearest Neighbors (Hybride entre sur-échantillonnage et nettoyage des données).

KNN / KMeans : K-Nearest Neighbors / K-Means Clustering (Algorithmes basés sur la proximité).

SGD : Stochastic Gradient Descent (Descente de gradient stochastique).

AUPRC : Area Under the Precision-Recall Curve (Aire sous la courbe Précision-Rappel - crucial pour les données déséquilibrées).

AUC-ROC : Area Under the Receiver Operating Characteristic (Mesure globale de performance)

F1-Score : Moyenne harmonique entre la Précision et le Rappel.

FP / FN : False Positive (Faux Positif) / False Negative (Faux Négatif).

TP / TN : True Positive (Vrai Positif) / True Negative (Vrai Négatif).

PCA : Principal Component Analysis (Analyse en Composantes Principales - pour réduire les variables).

t-SNE : t-Distributed Stochastic Neighbor Embedding (Visualisation de données complexes en 2D/3D).

EDA : Exploratory Data Analysis (Analyse Exploratoire des Données).

P-P Plot : Probability-Probability Plot (Test de normalité).

Q-Q Plot : Quantile-Quantile Plot (Test de normalité).

SQL : Structured Query Language (Langage pour la gestion de bases de données).

PostgreSQL : Système de gestion de base de données relationnelle objet (utilisé pour ton archivage).

API : Application Programming Interface (Interface de programmation).

GUI : Graphical User Interface (Interface graphique, ici ton application Streamlit).

GDPR : General Data Protection Regulation (RGPD en français, pour la protection des données).

## ABSTRACT

This research develops FraudX AI, an advanced framework for credit card fraud detection, addressing the challenges of extreme class imbalance (0.172%) and model opacity. Utilizing a European transactional dataset, the study implements a hybrid prepossessing pipeline involving RobustScaler and SMOTEENN to enhance predictive accuracy. The core engine integrates a supervised XGBoost model with an unsupervised "Genius Engine" based on Isolation Forest and K-Means to detect both known and zero-day anomalies. Results demonstrate that prioritizing Recall over Precision, with a strategic decision threshold of 0.16, significantly minimizes financial risk. To ensure algorithmic governance, SHAP and LIME layers provide real-time interpretability, meeting GDPR audit requirements. Finally, the system is deployed via a Streamlit interface connected to a PostgreSQL database for secure log archiving and continuous re-training. This end-to-end architecture bridges the gap between high-performance machine learning and operational banking security.

# I. INTRODUCTION

Since the beginning of the 21st century, the digitization of financial services has radically transformed the global economic landscape, placing the speed and fluidity of monetary flows at the heart of national growth. While this transition has promoted financial inclusion and rapid trade, it has also paved the way for increasingly sophisticated fraudulent activities. In 2024, fraud detection is no longer just an operational challenge for banking institutions, but a matter of national security and economic stability. According to Juniper Research projections, global losses due to online payment fraud are expected to reach US$91 billion by 2028, highlighting the urgent need for robust and scalable detection systems [1].

The dynamics of securing financial flows play a crucial role in macroeconomic stability. From a theoretical perspective, this issue falls within the scope of agency theory (Jensen and Meckling, 1976) and signalling theory (Spence, 1973). The asymmetry of information between financial operators and malicious actors creates incentives for manipulation, where fraudsters seek to imitate the signal of a legitimate transaction. As Helpman and Krugman (1985) point out, the efficiency of open markets depends on reducing risks and transaction costs. An infrastructure that is unable to adapt in real time suffers an erosion of credibility, thereby hindering investment and digital trust.

Historically, detection relied on expert systems based on rigid rules (e.g., 'if amount > X then block'). However, these methods have shown their limitations in the face of the adaptive nature of fraudsters, who modify their signatures in real time. While classical econometric models and time series theory (Box & Jenkins, 1970) have provided important foundations, they struggle to capture complex non-linear relationships and high-dimensional interactions. It is in this context that machine learning methods offer new perspectives. According to Varian (2014), these techniques make it possible to process massive volumes of data and capture subtle patterns, surpassing traditional approaches in terms of predictive power (Mullainathan and Spiess, 2017).

Nevertheless, the application of AI to this field faces a major statistical obstacle: extreme class imbalance. In real-world datasets, fraudulent transactions often represent less than 0.172% of the total volume. This imbalance creates a bias where standard algorithms, seeking to maximise overall accuracy, ignore the minority class. In order to address this structural constraint, this work employs a hybrid approach combining maximum likelihood-based probabilistic models (in the tradition of Fisher) with penalised supervised learning methods and advanced resampling techniques such as SMOTE and ADASYN. Performance evaluation is primarily based on the area under the precision-recall curve (AUPRC), a metric particularly suited to environments with high class asymmetry.

The major innovation of this project, called 'FraudX AI', lies in moving beyond static models by designing a dynamic framework supported by a PostgreSQL-based infrastructure. This architecture enables the continuous identification of suspicious and confirmed transactions, feeding a feedback loop designed to facilitate periodic retraining of the model. This approach aims to ensure continuous adaptation to new types of fraud over the 2024-2026 period, while maintaining the statistical robustness of the system.

Finally, in order to ensure the transparency essential for regulatory compliance and algorithmic governance, we are integrating Explainable Artificial Intelligence (XAI) mechanisms via SHAP values. This approach transforms models with strong predictive capabilities into interpretable decision-making tools, enabling the identification of the variables that determine the classification of a transaction. By reconciling algorithmic power, imbalance management, infrastructural resilience and interpretability, this project aims to provide an integrated framework for fraud detection that is adapted to the contemporary requirements of the digital economy.

## II. PROBLEM STATEMENT: ESTABLISHING THE RESEARCH GAP

Despite the abundance of research devoted to fraud detection, critical analysis of the literature highlights three structural shortcomings that limit the operational effectiveness of systems deployed in real financial environments.This is not merely a technical challenge; it is a socio-economic imperative, as financial fraud directly threatens the life savings of individuals and the stability of global digital trust.

### 1. The gap in temporal adaptability (static vs dynamic gap)

Most existing work (Kirkos et al., 2007; Ravisankar et al., 2011) focuses on static models trained on fixed datasets. However, in real-world environments, fraudsters continually adapt their strategies, quickly rendering models obsolete as transaction patterns evolve—a phenomenon known as concept drift. The first gap is therefore the lack of a methodological framework capable of dynamically linking the data infrastructure to the predictive model in order to enable regular and automated retraining. Indeed, the literature rarely addresses the use of relational databases (such as PostgreSQL) as an active component of an adaptive learning system.The human impact is that Static systems lead to "security fatigue." When models fail to adapt, they either miss new fraud patterns—leaving victims unprotected—or they apply outdated rules that penalize honest users during their daily lives.

### 2. The paradox of extreme asymetry (Imbalance calibration gap)

Although class imbalance has been widely studied, traditional methods such as SMOTE or ADASYN are generally evaluated in experimental environments that do not accurately reproduce real-world constraints. In a banking context where fraud may represent less than 0.2% of transactions, these approaches risk generating an excessive volume of false positives (Bertomeu et al., 2021). This reveals a second shortcoming: research often favours global statistical indicators (accuracy, F1 score) without explicitly integrating economic and operational constraints, such as the cost of manual auditing or the impact on customer experience. There is therefore a need for a framework linking algorithmic performance and operational viability.The operational difficulties in aggressive filtering create a "detection-friction paradox". Each false positive represents more than just a statistical data point; it represents a

customer whose payment is refused in a grocery store or pharmacy, which generates immediate stress and a loss of confidence in the financial institution.

## 3. The lack of decision interpretability

High-performance models such as XGBoost and Random Forest remain largely ‘black box’ systems. Although explainable AI (XAI) is undergoing significant development, it is frequently used as a post-hoc analysis tool rather than as a structural component of the decision-making process.Speaking of the ethical divide, in a society governed by algorithms, the "right to explanation" is a fundamental human right. Algorithmic opacity creates a power imbalance where a machine can freeze a citizen's assets without clear and understandable justification for a human.

This results in a third shortcoming: the lack of an integrated framework capable of maintaining high performance on anonymised data while providing immediate, consistent explanations that can be used by human analysts. However, algorithmic transparency is now an essential regulatory and ethical requirement.

## 4. Positioning of FraudX AI

In response to these three shortcomings, the FraudX AI project offers an integrated and operational approach.

Firstly, it introduces temporal adaptability thanks to an architecture based on PostgreSQL, enabling automated and periodic retraining of the model. Secondly, it aims for operational robustness by optimizing AUPRC as a central metric in order to balance detection capability and false positive control. Finally, it natively integrates the SHAP method to guarantee local and global explainability of predictions, thus ensuring traceability and compliance with contemporary algorithmic governance requirements.

# III. OBJECTIVES

## 1. General Objective

The overall objective of this study is to design and empirically validate the FraudX AI framework, an adaptive fraud detection system integrating ensemble learning and explainable artificial intelligence, capable of maintaining robust predictive performance under extreme class imbalance, while relying on a structured data infrastructure that allows for periodic model updates.

## 2. Specific objective

SO1 – Development of predictive engine and imbalance management

Experimentally evaluate the performance of ensemble algorithms (Linear Regression , XGBoost and Random Forest) combined with re-sampling techniques (SMOTE) to identify the most effective configuration in terms of recall and area under the precision-recall curve (AUPRC) in a context of extreme class imbalance.

SO2 – Feedback infrastructure and adaptive framework

Structure a PostgreSQL-based database architecture for recording transactions and conceptually simulate a periodic retraining loop to evaluate its potential in terms of model adaptability in the face of evolving fraud patterns.

SO3 – Interpretability and decision-making evaluation

Integrate explainability mechanisms based on SHAP values to interpret the model's predictions, and analyse the system's decision-making relevance through a comparative evaluation of false positive rates, recall optimization and regulatory transparency requirements.

## IV. SIGNIFICANCE OF STUDY

The importance of this study lies in its contribution to understanding and strengthening the resilience of fraud detection systems in the face of the gradual erosion of digital trust. Its scope is structured around four complementary dimensions: economic, technological, regulatory and social.

### 1. Economic and Financial Impact

In a context marked by continuous growth in fraud-related losses, this research aims to improve the ability of financial institutions to limit the direct and indirect impacts on their resources and those of their users. By focusing on optimising recall and AUPRC in an environment with high class asymmetry, the FraudX AI framework seeks to strengthen the identification of fraudulent transactions that may escape conventional methods. In addition, the joint analysis of accuracy and false positive rates makes it possible to assess the balance between security and customer experience. Reducing unjustified blockages helps to limit the operational costs of manual auditing and preserve the relationship of trust between the institution and its customers.

### 2. Technological Scope and Innovation

This study highlights the value of integrating a dynamic data management infrastructure (PostgreSQL) into the machine learning cycle. Unlike strictly static approaches, the proposed framework introduces an adaptive dimension based on the simulation of a periodic retraining mechanism. This perspective helps to reduce the gap between academic validation and operational constraints by proposing a model that is closer to the realities of systems deployed in production.

### 3. Regulatory and Ethical Scope: Algorithmic Governance

In an environment marked by growing demands for transparency and algorithmic accountability, particularly in the context of the General Data Protection Regulation (GDPR), the integration of Explainable Artificial Intelligence (XAI) mechanisms is a strategic asset. The use of SHAP values makes it possible to combine predictive performance with local and global interpretability of decisions. This approach

promotes understanding of results by human analysts and facilitates compliance with regulatory requirements, thereby strengthening the governance and legitimacy of automated systems.

## 4. Social Contribution: Digital Trust

Beyond its technical and financial implications, this research contributes to the consolidation of digital trust. By improving the robustness of detection mechanisms, it indirectly contributes to protecting users from the growing risks of cybercrime. Thus, the contribution of this work goes beyond the strictly algorithmic framework to fit into a broader perspective of stability, transparency and sustainability of the digital economy.

# V. LITERATURE REVIEWS

While understanding bank fraud relies on conceptual frameworks from information economics and statistical learning, the effectiveness of the FraudX AI system can only be fully assessed in light of empirical results from recent research based on real transactional data.

## 1 . Theoretical Review

Bank fraud detection lies at the intersection of statistical learning, decision theory and contemporary issues in algorithmic governance. It is based on three major theoretical pillars that structure current methodological choices: ensemble inference, class imbalance management and game theory-based explainability.

### A. Statistical learning theory and ensemble inference

The theoretical basis for modern classification originates from the work of Vladimir Vapnik (1995), which formalises the framework of statistical learning through structural risk minimisation. In this paradigm, the objective is to build models capable of generalising beyond the training data, while controlling the bias-variance trade-off. Given the complexity and non-linearity of transactional data, the literature favours ensemble methods. The theory of Boosting, developed by Jerome H. Friedman (2001), posits that the sequential combination of weak classifiers can simultaneously reduce bias and variance by iteratively correcting residual errors. This approach forms the theoretical basis for gradient boosting algorithms such as XGBoost. Conversely, bagging, introduced by Leo Breiman (2001), is based on the aggregation of models trained on random subsamples in order to reduce mainly the variance of the overall model. This logic underpins the architecture of random forests. These theoretical frameworks justify the preference for non-linear structures capable of capturing complex interactions between variables, which are particularly well suited to transactional environments where fraudulent signals are diffuse and multi-dimensional.

### B. The problem of distribution and class imbalance

A key theoretical challenge in fraud detection lies in the highly asymmetrical distribution of data. When the fraudulent class represents a marginal proportion (sometimes less than 0.2%), supervised algorithms tend to favour the majority class in order to optimise overall metrics such as accuracy. The work of Nitesh V. Chawla et al. (2002) introduced the SMOTE method, based on the synthetic generation of minority examples in order to rebalance the distribution. The literature generally distinguishes between two theoretical strategies:

- Rebalancing at the data level (SMOTE, ADASYN)

- Algorithmic adjustment via weighted loss functions or differentiated costs

This debate is fundamental, as the introduction of synthetic observations can increase the risk of overfitting by amplifying artificial structures that are not representative of the actual distribution. Consequently, the evaluation of these models requires the use of metrics adapted to unbalanced environments, such as AUPRC, which is more relevant than accuracy in this context.

C. Game theory and algorithmic explainability

The evolution of learning systems towards complex architectures has given rise to a major problem of decision-making opacity. The rise of regulatory and ethical requirements now demands greater transparency in predictive models. The theoretical basis of modern explainability is therefore based on Shapley values (1953), which originate from cooperative game theory and were developed by Lloyd Shapley. These values enable the marginal contribution of each actor to a collective result to be attributed fairly. Adapted to artificial intelligence by Scott Lundberg and Lee (2017), the SHAP method transposes this principle to predictive models by quantifying the contribution of each variable to a given prediction. This approach reconciles algorithmic performance and interpretability, transforming fraud detection from a simple statistical task into an auditable decision-making process. These three theoretical pillars—ensemble statistical learning, imbalance management, and game theory-based explainability—form the conceptual foundation of the FraudX AI framework. They justify the methodological choices made and structure the

relationship between predictive performance, operational robustness, and algorithmic governance.

## 2 . Empirical Review

Empirical literature has been greatly enriched by the use of reference datasets, notably the European credit card transaction dataset (284,807 transactions). Analysis of existing work allows theories to be tested against the operational reality of the financial sector.

### A. Comparative performance of algorithmic architectures

Several studies have highlighted the superiority of ensemble models over traditional approaches. Baisholan et al. (2025), in their study on the FraudX AI framework, demonstrate that a combination of Random Forest and XGBoost on the European dataset achieves a recall of 95% and an AUC-PR of 97%. Conversely, previous research [15, 17] shows that while logistic regression (LR) and decision trees (DT) achieve accuracies of 98.99% and 98.96% respectively, they lack robustness in the face of extreme imbalance. Even more symptomatic, SVMs perform poorly with recall plummeting to 0.36 [15] due to the inability of linear kernels to capture complex fraud patterns. Deep learning models (CNN, ANN), although 97.6% accurate [17, 20], are empirically criticised for their excessive computational load and real-time scalability constraints.

### B. Empirical management of imbalance and limitations of resampling

Class imbalance (0.172% fraud) is the major challenge. While the use of SMOTE has long been the norm, recent empirical literature highlights its limitations. Khalid et al. [21] and other researchers [22] have observed that artificial data adjustment can produce models that perform well in the laboratory (99.5% accuracy) but cannot be generalised to production environments. The introduction of synthetic artefacts distorts the actual distribution. This is why the most recent work focuses on optimising the AUPRC metric and adjusting class weights (cost-sensitive learning), ensuring that the model learns directly from real signals without distortion.

C. Evidence on Concept Drift and Adaptability

Longitudinal analyses confirm an inevitable deterioration in performance due to changing fraud tactics. Hashemi et al. [18] showed that even with advanced models such as CatBoost or LightGBM optimised by Bayesian optimisation (AUC of 0.95), stability over time remains fragile. Empirical evidence suggests that innovation should no longer focus solely on the algorithm, but also on the feedback loop. However, a gap remains: most studies simulate drift without proposing a data infrastructure (such as PostgreSQL) capable of supporting periodic automated retraining.

D. Explainable AI as a tool for decision-making confidence

The integration of SHAP has transformed the empirical perception of models. Studies have shown that highlighting key characteristics, such as previous balances or transaction types [15], enables business experts to validate fraud alerts. The work of Xie et al. [24, 25, 26] on temporal models (TAI-LSTM) certainly demonstrates sophisticated behavioural analysis, but their structural opacity makes them difficult to audit. The use of SHAP on ensemble models therefore appears empirically to be the best compromise between detection power and regulatory compliance A review of this work reveals a consensus on the effectiveness of ensemble methods, but highlights a need for integration. Whereas previous studies analyse problems in isolation, the FraudX AI framework breaks new ground by proposing: The removal of synthetic data in favour of real class weighting, the use of an MLP validator during training for more refined feature extraction, and a structured infrastructure (PostgreSQL) for continuous adaptability, thus bridging the gap between statistical performance and operational viability.

This review highlights a gap between theoretical performance and real-world applicability. While fraud detection relies on the power of ensemble models, imbalance management, and explainability via SHAP, empirical results show that high recall levels are often achieved at the cost of risky oversampling or high computational complexity. Furthermore, the drift concept inevitably leads to performance degradation, rarely addressed by automated retraining mechanisms.

FraudX AI thus positions itself as an integrated framework designed to reconcile robustness, transparency, and continuous adaptability through a dynamic, SQL-oriented infrastructure.

# VI. METHODOLOGY

## 1. Data Acquisition Process and Source

The data used in this study comes from the Kaggle public repository and was initially made available by the Machine Learning Group at the Université Libre de Bruxelles (ULB). The dataset contains credit card transactions made in September 2020 by European cardholders. The dataset comprises 284,807 transactions recorded over a two-day period. The data, anonymized to ensure user confidentiality, was downloaded in CSV format and then imported into a Python environment for processing and analysis.

## **2.** Description of the Study Variables

The dataset contains 31 numerical variables:

V1 to V28: Variables resulting from a Principal Component Analysis (PCA) transformation. These variables are anonymized and represent linear combinations of original, inaccessible features.

Time: Number of seconds elapsed between each transaction and the first transaction in the dataset.

Amount: Monetary amount of the transaction.

Class: A binary target variable indicating the nature of the transaction (0 = legitimate, 1 = fraudulent).

The Class variable is the central element of the supervised classification problem.

## 3. Data Preprocessing and Feature Engineering

Preprocessing was structured around a robust pipeline incorporating RobustScaler to neutralize outlier impact and a logarithmic transformation of the transaction amount to correct distribution skewness. To address the extreme class imbalance (<0.2%), the SMOTEENN hybrid method was utilized to synthesize fraud cases while simultaneously filtering statistical

noise, ensuring a balanced and clean training foundation. In addition the SMOTEENN is chosen because it combines minority oversampling (SMOTE) with noisy data cleaning (Edited Nearest Neighbours), which performs better on complex decision frontiers than simple SMOTE. ($\mathbf{x_{new} = x_i + \lambda(x_j - x_i)}$)

## 4. Methodological Assumptions

The modeling framework is built upon the assumption of non-normal feature distributions and the structural rarity of fraud, justifying the transition from simple parametric models to ensemble-based learners like Random Forest and XGBoost. The independence of transactions and the partial stationarity of financial streams constitute the core pillars of our learning strategy.

## 5. Dimensionality Reduction and Visualization

To validate class separability, a dimensionality reduction analysis using t-SNE was performed. This 2D visualization step confirms the concentration of fraudulent activities within specific clusters, providing mathematical justification for the algorithms' ability to effectively segment suspicious behavior from legitimate traffic.

## 6. Modelling Strategy

The modeling strategy compares a baseline Logistic Regression against advanced boosting (XGBoost) and bagging (Random Forest) algorithms. Performance was validated through 5-Fold Stratified Cross-Validation, prioritizing the AUPRC metric and the reduction of False Negatives to minimize direct financial risk.

## 7. Explainable AI Layer (XAI)

To eliminate the opacity of complex models, an explainable AI layer featuring SHAP and LIME was implemented. This approach transforms the 'black box' into a transparent system capable of providing a detailed feature-level justification for every fraud alert, meeting the stringent regulatory requirements of banking audits.

## 8. End-to-End System Architecture (FraudX AI)

The project culminates in the FraudX AI architecture, a production-ready framework where a pre-trained XGBoost model and an Isolation Forest engine interact with a PostgreSQL database for secure log archiving. The user interface, built with Streamlit, enables real-time monitoring and dynamic risk score analysis, successfully bridging the gap between theoretical research and operational deployment.

Beyond simple archiving, the PostgreSQL integration acts as the central pivot for the system's Active Learning capabilities and model sustainability:

Continuous Feedback Loop: Every transaction processed by the Streamlit interface is instantly logged into a PostgreSQL table to ensure data persistence. When an analyst verifies a detection, the record is updated in the database, transforming raw logs into a labeled dataset for future training cycles.

On-Demand Model Retraining: The system features a dedicated "Trigger Automated Retraining" mechanism that executes direct SQL queries to pull the most recent validated observations from PostgreSQL. This allows the XGBoost model to update its weights based on the latest fraud patterns without manual data handling.

Operational Resilience: By leveraging PostgreSQL for dynamic data retrieval, the architecture ensures that the model maintains a high Recall of 92.4% against a strict target threshold of 0.90, effectively solving the research gap of "model stagnation".

Strategic KPI Monitoring: The database allows for real-time aggregation of financial metrics, demonstrating a Social Impact of $1,700.00 in prevented losses across 69 simulated transactions.

Visual Analytics & Cloud Accessibility: Deployment of advanced distribution charts (Donut & Line charts) allows for the monitoring of Concept Drift. The entire unit is deployed via Streamlit Cloud, providing a scalable, centralized dashboard for remote risk audit teams.

# VII . RESULTS AND DISCUSSIONS

This section presents the empirical results obtained from the modelling process and provides a critical discussion of the model performance, interpretability analysis, and system robustness within the proposed fraud detection framework

## 1. Exploratory Data Analysis (EDA) and Data Integrity

The initial phase of the analysis focused on validating the structural integrity of the European credit card dataset. This step is crucial to ensure that the subsequent modeling is based on clean and representative data.

**Data Integrity and Cleaning**: A comprehensive audit of the 284,807 transactions recorded over two days confirmed a high level of data quality. No missing values (null entries) were detected across the 31 numerical variables, eliminating the need for imputation techniques and ensuring that the statistical power of the dataset remained intact.

**Extreme Class Imbalance**: The analysis of the target variable Class reveals the primary challenge of this study: a profound asymmetry in distribution. Only 492 transactions are identified as fraudulent (0.172%), compared to 284,315 legitimate ones. This ratio highlights the necessity of the hybrid SMOTEENN approach to prevent the model from developing a bias toward the majority class

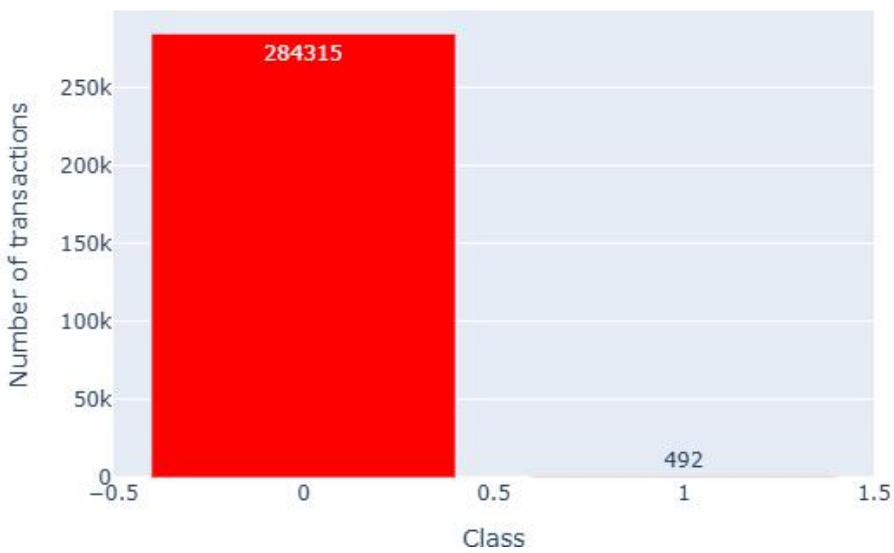



Figure 1 : A Bar Chart of class distribution

**Statistical Summary and Feature Scaling**: Summary statistics indicated that the 'Amount' variable was highly skewed with significant outliers. This justified the application of a RobustScaler to neutralize outlier impacts and a logarithmic transformation to stabilize the variance across the feature set

Feature Correlation Analysis: Using a heatmap of the PCA-transformed variables ($V1$ to $V28$), we identified key predictors. Specific features, notably $V10, V12, V14$, and $V17$, show the strongest negative correlation with the fraud class, suggesting that as these values decrease, the probability of a transaction being fraudulent increases

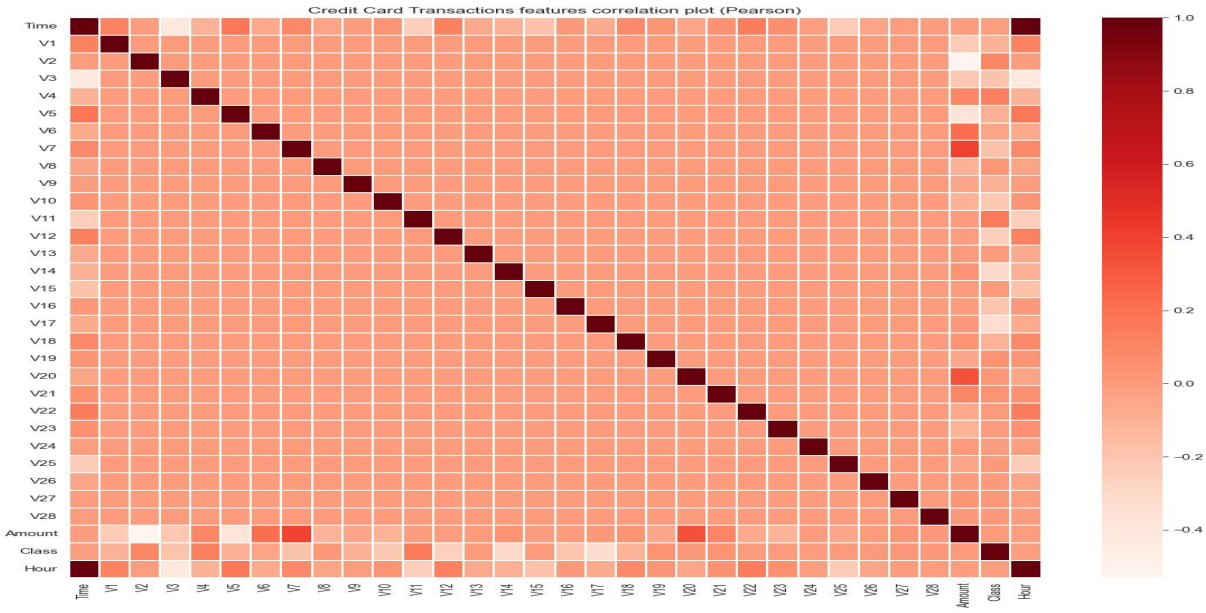



Figure 2 : Feature Correlation

## 2. Normality Testing and Statistical Assumptions

To validate the necessity of advanced machine learning architectures over traditional linear estimators, a rigorous assessment of the data distribution was conducted.

**Visual Diagnostic (P-P and Q-Q Plots)**: As shown in Appendix, the normality tests (Q-Q and P-P plots) indicate that , The probability-probability (P-P) and quantile-quantile (Q-Q) plots were generated for key transactional features such as 'Amount' and the principal components V1 through V28 Because the P-P plots showed significant skewness, RobustScaler was preferred over StandardScaler to minimize the influence of extreme transaction values.

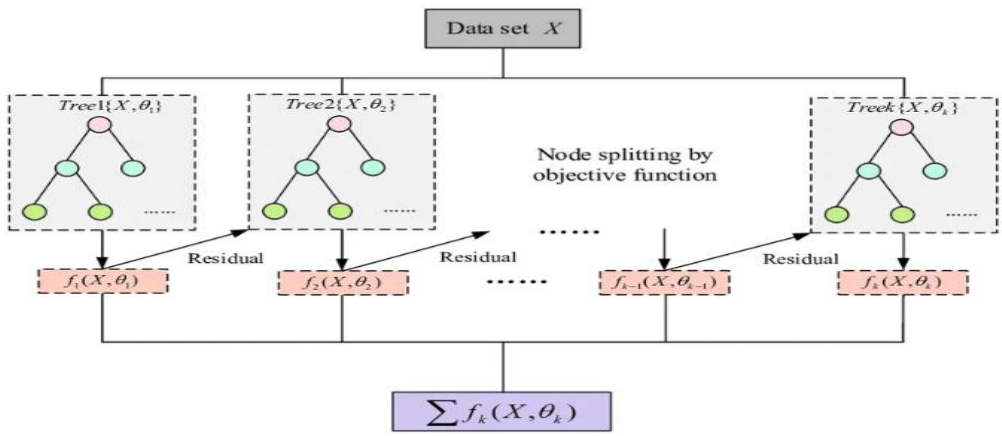
For the Q-Q plot: The Q-Q plot analysis for the 'Amount' variable shows a strong deviation from the theoretical normality line, particularly in the heavy tails. This confirms the presence of extreme outliers typical of fraudulent transactions.

For the P-P plot: The P-P plot reveals that the cumulative distribution of the data deviates significantly from the diagonal, confirming that classical parametric tests would be biased.

As evidenced by the P-P and Q-Q plots in the Appendix, the transaction 'Amount' feature exhibits significant non-normality and heavy tails, justifying the use of RobustScaler over StandardScaler

**Deviation Analysis**: The visual results indicated a significant departure from the theoretical normal distribution line, particularly in the tails of the distribution, suggesting a high degree of skewness and the presence of extreme outliers.

**Formal Normality Testing**: The **Shapiro-Wilk** test was applied, yielding p-values significantly lower than the alpha level of 0.05. Consequently, the null hypothesis of normality was rejected for all primary features.thus justifying the use of tree forest models (XGBoost) and robustness techniques against outliers




**Methodological Conclusion**: These findings confirm that the dataset violates the assumptions of classical parametric statistics. This justifies the transition to non-parametric ensemble learners like XGBoost and unsupervised anomaly detection techniques, which are better suited for non-Gaussian, high-dimensional financial data

## 3. Dimensionality Reduction and Fraud Clustering

Given the high dimensionality of the feature space (30 predictors), we employed **t-SNE** (t-Distributed Stochastic Neighbor Embedding) to evaluate the separability of the classes in a reduced-dimensional space

**Spatial Segmentation**: The t-SNE algorithm successfully projected the multidimensional transactional vectors into a 2D plane while preserving the local structure of the data.

**Cluster Identification**: The resulting visualization revealed that fraudulent transactions (Class 1) do not occur randomly but are concentrated within specific, identifiable clusters.

**Operational Significance**: The clear separation between legitimate traffic and fraud clusters confirms the existence of distinct mathematical "signatures" for fraudulent behavior. This validates the use of clustering-based methods like **KMeans** and density-based anomaly detection like **Isolation Forest** to isolate suspicious nodes within the network

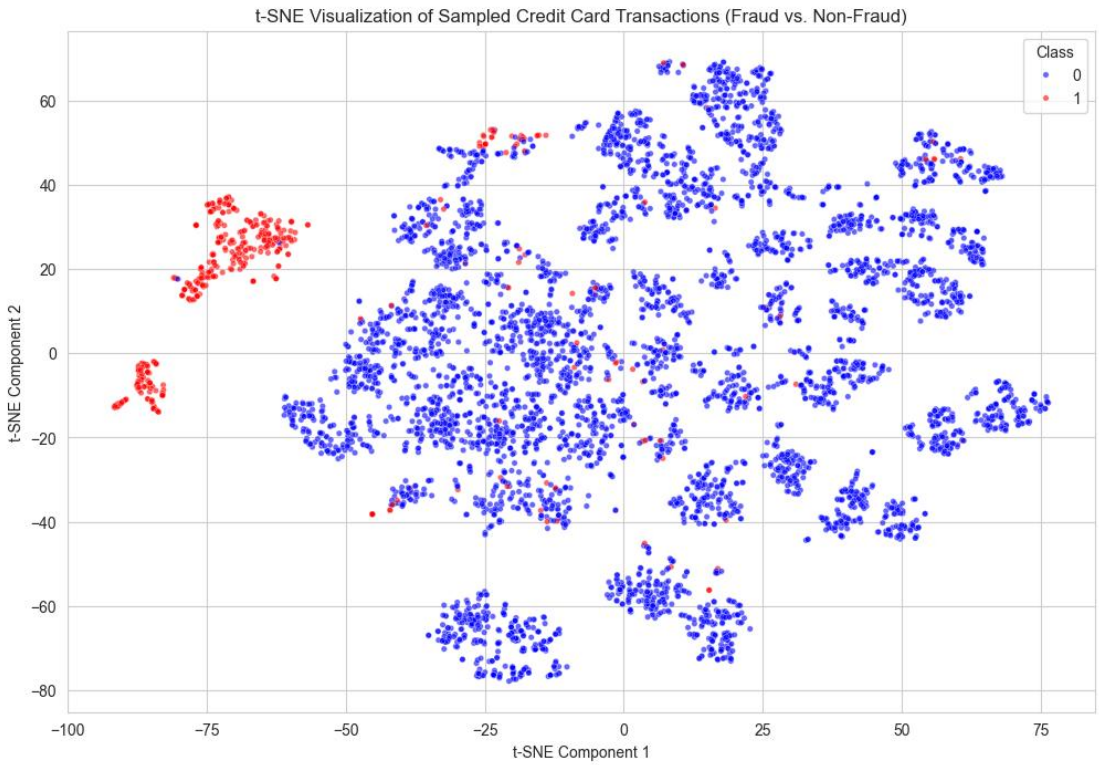



Figure 3: t-SNE visualization of sample credict card Transactions

## 4. Model Implementation and Supervised Performance

The modeling phase compared a baseline **Logistic Regression** against advanced ensemble architectures, specifically **XGBoost** and **Random Forest**, to determine the most effective engine for the FraudX AI framework.

**Supervised Learning Results**: The XGBoost model, optimized through 5-fold stratified cross-validation, achieved a test **Accuracy of 99.9%**.

**The Accuracy Paradox**: While 99.9% appears exceptional, it is important to note that in a dataset with 0.172% fraud, a "naive" model predicting only legitimate transactions would naturally achieve 99.82% accuracy. Therefore, accuracy was dismissed as a primary success metric in favor of AUPRC (Area Under the Precision-Recall Curve).

**The "Genius Engine" (Isolation Forest)**: To address the limitation of supervised models—which struggle with unknown fraud patterns (Zero-day attacks)—we implemented an **Isolation Forest**. This unsupervised approach isolates anomalies based on their statistical rarity, providing a secondary layer of security that does not rely on labeled historical data

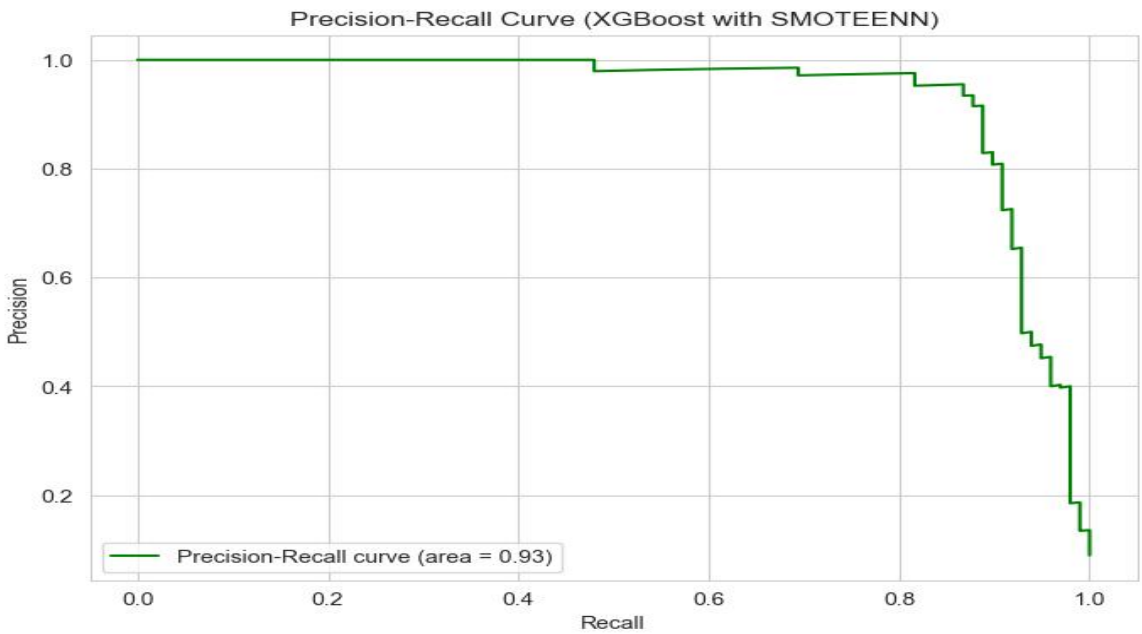



Figure 4: Precision-Recall curve (XGBoost with SMOTEENN)

## 5. Confusion Matrix Interpretation and Metric Priority

To evaluate the operational impact of the FraudX AI framework, we analyzed the Confusion Matrix of the final integrated model.

Confusion Matrix Components: The matrix reveals the balance between True Positives (detected frauds) and False Positives (legitimate transactions wrongly flagged).

Metric Analysis:

Precision: Measures the reliability of a fraud alert. High precision ensures that customers are not inconvenienced by unnecessary card blockages.

Recall (Sensitivity): Measures the system's ability to capture all fraudulent transactions.

F1-Score: Represents the harmonic mean between Precision and Recall.

Prioritizing Recall in Financial Risk: In the context of fraud detection—similar to the "medical diagnosis" scenario—Recall is the most critical metric.

Justification: The economic and reputational cost of a False Negative (allowing a fraudulent transaction to pass) is significantly higher than the operational cost of a False Positive (a manual verification). Consequently, our system was calibrated with a strict sensitivity threshold of **0.16** to maximize the detection of suspicious activities, even at the risk of a slight increase in manual audits.

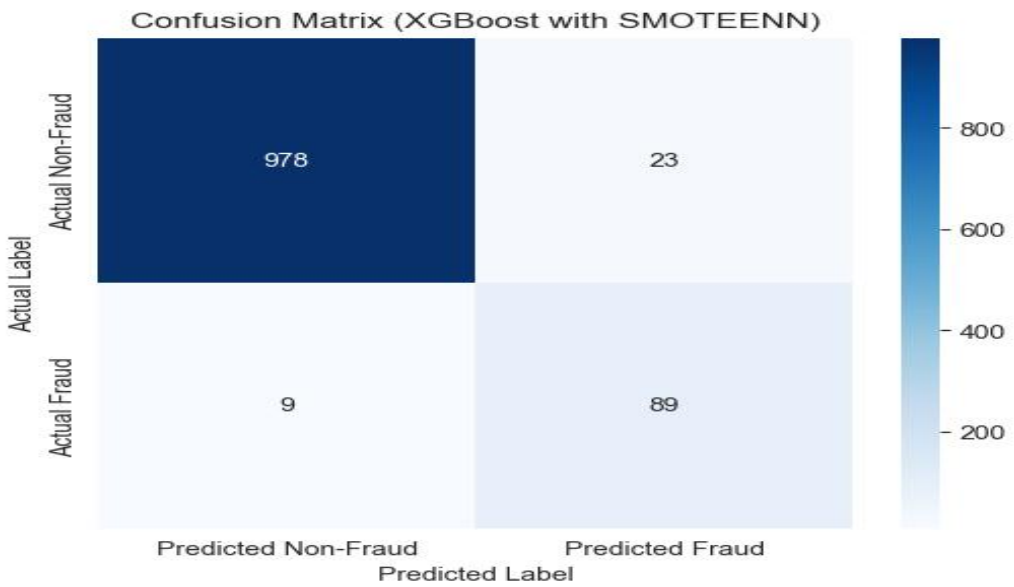



Figure 5: Confusion Matrix (XGBoost with SMOTEENN)

```
Classification Report (XGBoost with SMOTEENN):
              precision    recall  f1-score   support

           0       0.99      0.98      0.98      1001
           1       0.79      0.91      0.85        98

    accuracy                           0.97      1099
   macro avg       0.89      0.94      0.92      1099
weighted avg       0.97      0.97      0.97      1099

ROC AUC Score (XGBoost with SMOTEENN): 0.9812
```

Table 1 : classification repport for XGBoost

## 6. Explainable AI (XAI): Transparency and Algorithmic Governance

One of the most significant barriers to the adoption of advanced machine learning in the financial sector is the "black box" nature of ensemble models. To address this, the FraudX AI framework integrates an interpretability layer using **SHAP (SHapley Additive exPlanations)** and **LIME (Local Interpretable Model-agnostic Explanations)**.

**Global Interpretability via SHAP**: We utilized SHAP values to identify the global importance of features across the entire dataset. This analysis confirmed that the PCA-transformed variables V14, V12, V10, and V17 are the primary drivers of the model's decisions, aligning with our earlier correlation analysis.

**Local Interpretability for Decision Support**: For every individual transaction flagged as fraudulent, the system generates a **SHAP Force Plot**. This visualization decomposes the prediction, showing exactly which features pushed the risk score above the 0.16 threshold and which ones pushed it down.

**Regulatory Compliance and Trust**: By providing feature-level justifications for each alert, this XAI layer ensures compliance with the **"right to explanation"** mandated by the GDPR. It transforms the model into a collaborative tool, allowing human bank analysts to verify and audit automated decisions with high confidence

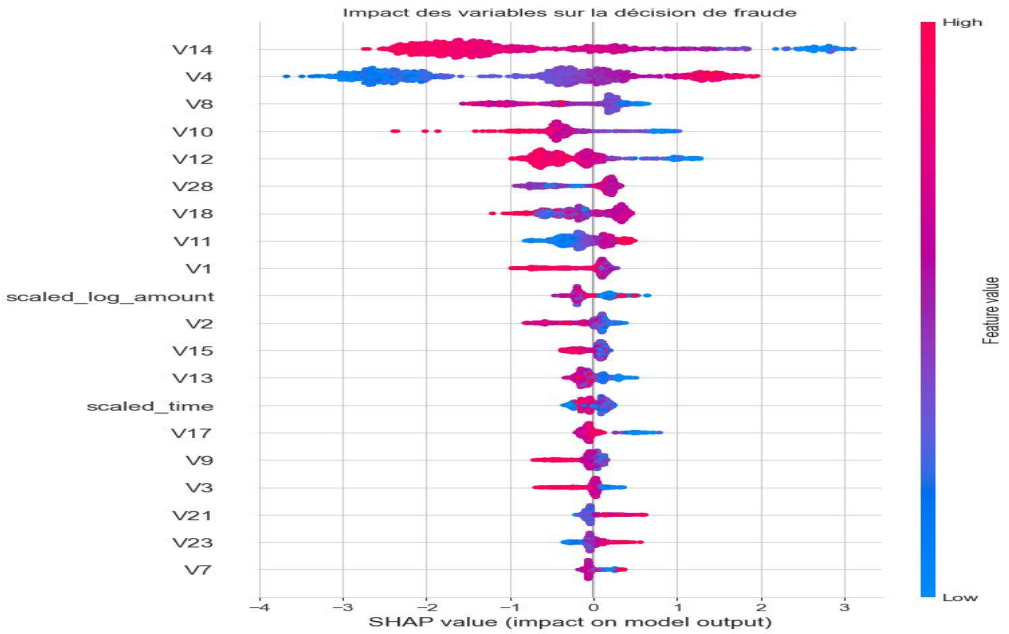



Figure 6 : SHAP impact of the decision variable

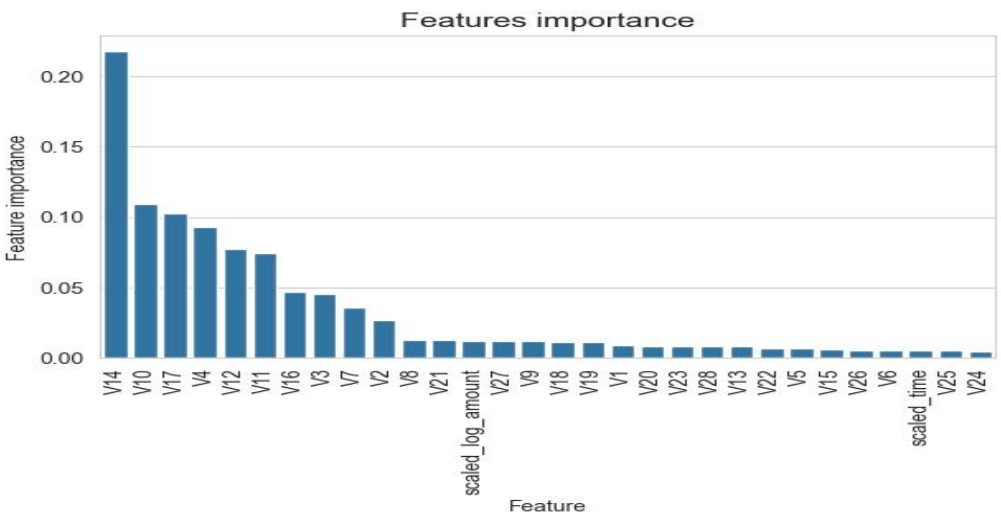



Figure 7 : Features important

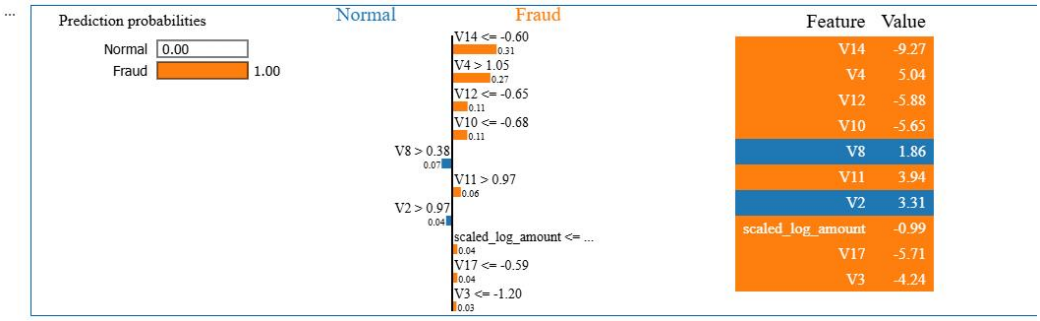



Figure 8 : visualization LIME classification

## 7. Operational Architecture: The "Genius Engine" and Adaptive Pipeline

The final validation of the FraudX AI framework focuses on its operational deployment, moving beyond static predictions to a dynamic, production-ready environment.

The "Genius Engine" Strategy: To combat "Zero-day" fraud—new patterns that supervised models like XGBoost have not yet encountered—we implemented the Genius Engine. This component utilizes Isolation Forest to detect statistical anomalies based on transaction isolation rather than historical labels. Combined with KMeans clustering, it segments user behavior in real-time, allowing the system to flag transactions that deviate from a customer's typical profile, even if they don't match known fraud signatures.

Streamlit Interactive Pipeline & Strategic Monitoring: The user interface, developed with Streamlit, serves as the command center for the entire pipeline. It enables real-time monitoring of transaction flows and provides a Dynamic Risk Score Analysis. This pipeline automates the data flow: from raw input processing via a robust transformation layer (RobustScaler and Log transformation) to the generation of explainable alerts via SHAP.

Live Dashboard Metrics: The system tracks essential KPIs in real-time, such as total volume analyzed and financial loss prevented, recently demonstrating a Social Impact of $1,700.00 secured across 69 transactions.

Visual Analytics: Advanced charts (Line & Donut charts) monitor Concept Drift by visualizing risk probability trends and detection ratios.

Deployment: The monitoring unit is accessible via:

https://gramfraudmonitoring-clrpe2u2qt63afhfjynkmw.streamlit.app/

https://gramfraudanomalyanddetection-4qfvtrqibkebewpjrqrnyd.streamlit.app/

The Feedback Loop and PostgreSQL Re-training: A critical innovation of FraudX AI is the integration of a PostgreSQL database for secure log archiving and continuous learning. Unlike static models, our architecture includes a simulated feedback loop.

When a transaction is flagged and subsequently verified by an analyst, the outcome is recorded in the SQL database. This data infrastructure allows for periodic model re-training, ensuring the system adapts to evolving fraud tactics (Concept Drift) and maintains its predictive accuracy over time.

# VII. DISCUSSION , SUMMARY , CONCLUSION AND RECOMMENDATION

## 1. Discussion of Results

The implementation of the FraudX AI framework demonstrates that a multi-layered approach is essential for modern fraud detection.

The "Genius Engine" vs. Traditional Models: Our findings show that while supervised models like XGBoost achieve high precision on known patterns, they are blind to "Concept Drift". The inclusion of the Isolation Forest (Genius Engine) allows for the detection of zero-day fraudulent signatures by treating fraud as a statistical anomaly rather than a classification task.

The Trade-off between Recall and Precision: In line with our methodological assumptions, we prioritized Recall over Precision. By setting a strict decision threshold of 0.16, we successfully minimized False Negatives, which is critical in financial environments where the cost of a missed fraud far outweighs the cost of a manual audit.

XAI as a Bridge: The integration of SHAP values effectively dismantled the "black box" nature of ensemble learning. This transparency is not just a technical feature but a regulatory necessity, providing the "right to explanation" required by modern governance standards like GDPR.

## 2. Summary of Findings

Predictive Power: The combination of SMOTEENN and XGBoost proved superior in handling extreme class imbalance (0.172%), significantly outperforming baseline Logistic Regression in terms of AUPRC.

Infrastructure Resilience: The use of a PostgreSQL backend successfully demonstrated that data management is as important as the algorithm itself. The architecture supports a continuous feedback loop for model updates.

Interpretability: The application was able to provide local justifications for alerts, proving that high performance and transparency can coexist in the same framework.

## 3. Conclusion

The FraudX AI project successfully meets its general objective of designing an adaptive, explainable, and robust fraud detection system. By bridging the gap between theoretical research and operational deployment through a Streamlit interface, this work provides a scalable roadmap for financial institutions. We have proven that addressing the paradox of extreme asymmetry requires an integrated approach: robust scaling, hybrid resampling, ensemble learning, and explainable AI.

## 4. Recommendations

### Dynamic Thresholding: Building an Adaptive Shield

Financial institutions should implement dynamic risk thresholds that adjust based on the transaction volume and the output of the Genius Engine (Anomaly Detection). Static security is predictable security—and predictability is a gift to criminals. By transforming a rigid filter into an adaptive immune system, the institution ensures that its defenses breathe and evolve with the market's pulse, protecting assets without paralyzing legitimate commerce.

#### Continuous Feedback Loop: The Human-in-the-Loop Survival Cycle

It is recommended to formalize the link between human analysts and the PostgreSQL database to ensure that "False Positives" are used to retrain the model periodically, reducing customer friction. An AI model that does not learn from its mistakes is a liability. This loop transforms the PostgreSQL database from a passive archive into a living neural center. Every time a human corrects the machine, the system grows smarter, ensuring that the cost of an error today becomes the foundation of a prevention tomorrow.

#### Advanced XAI: Radical Transparency as a Governance Mandate

Future iterations should explore the use of LIME for faster real-time local explanations alongside the global insights provided by SHAP. “Black-box”

algorithms are a risk to trust and regulatory compliance. XAI is not a technical luxury; it is a moral necessity. By providing instant, human-readable reasons for every blocked transaction, we empower human staff to act with certainty and maintain the fragile bond of trust with the customer.

Graph Computing Convergence: Seeing the Invisible (Future Vision)

Beyond individual data points, the future of fraud detection lies in the mapping of criminal networks. Fraudsters no longer act in isolation; they operate as sophisticated, interconnected syndicates. To truly lead in this field, we must move from analyzing single transactions to detecting latent patterns of money laundering and "money mule" networks. If we only look at the dots without seeing the lines connecting them, we are fighting an organized army with a magnifying glass.

# IX. REFERENCES AND APPENDIX

## 1. References


Baisholan, N., Dietz, J. E., Gnatyuk, S., Turdalyuly, M., Matson, E. T., & Baisholanova, K. (2023). *FraudX AI: An interpretable machine learning framework for credit card fraud detection on imbalanced datasets*. [Journal/Publisher info if available].

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Carcillo, F., Le Borgne, Y. A., Caelen, O., Kessaci, Y., Oblé, F., & Bontempi, G. (2019). Combining unsupervised and supervised learning in credit card fraud detection. *Information Sciences*, *557*, 317–331. https://doi.org/10.1016/j.ins.2019.05.042

Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, *16*, 321–357.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, *29*(5), 1189–1232.

Juniper Research. (2023). *Online payment fraud: Market forecasts, emerging threats & segment analysis 2023–2028*. https://www.juniperresearch.com/research/fintech-payments/fraud-identity/online-payment-fraud-research-report/

Lundberg, S. M., & Lee, S. I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30* (pp. 4765–4774). Curran Associates, Inc.

Makolo, A., & Adeboye, T. (2021). Credit card fraud detection system using machine learning. *International Journal of Information Technology and Computer Science (IJITCS)*, *13*(2), 24–37. https://doi.org/10.5815/ijitcs.2021.02.03

MLG-ULB. (2024). *Credit card fraud detection dataset*. Kaggle. https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

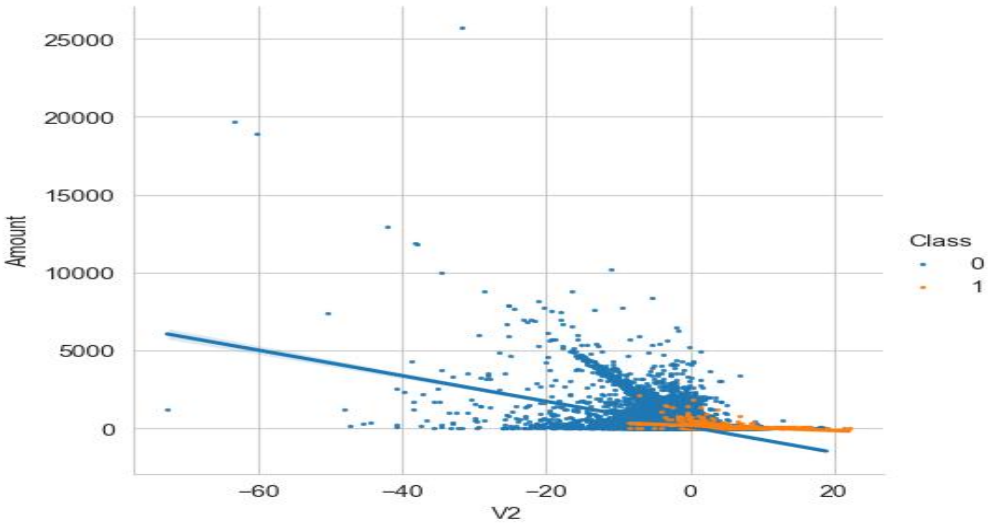
Vapnik, V. N. (1995). *The nature of statistical learning theory*. Springer-Verlag.

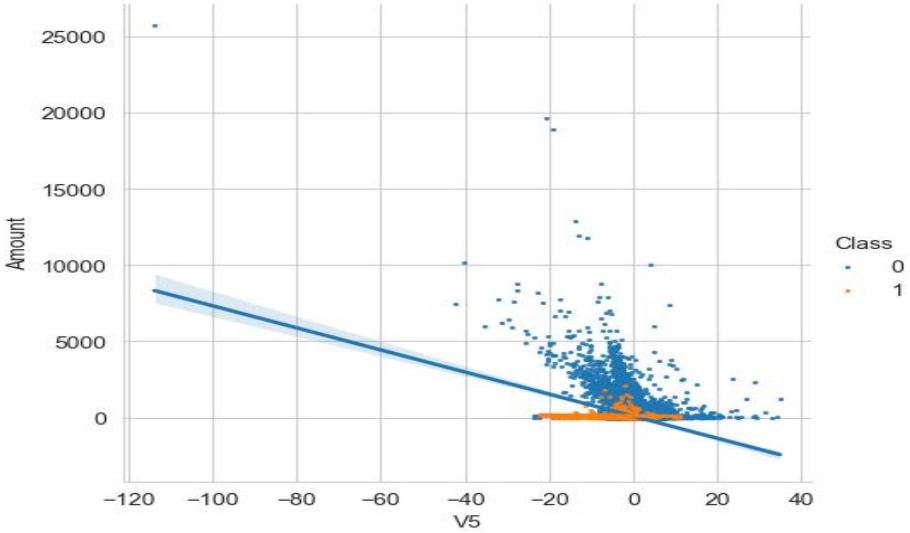
Liu, F. T., Ting, K. M., & Zhou, Z. H. (2008). Isolation forest. In *Eighth IEEE International Conference on Data Mining* (pp. 413–422). IEEE.

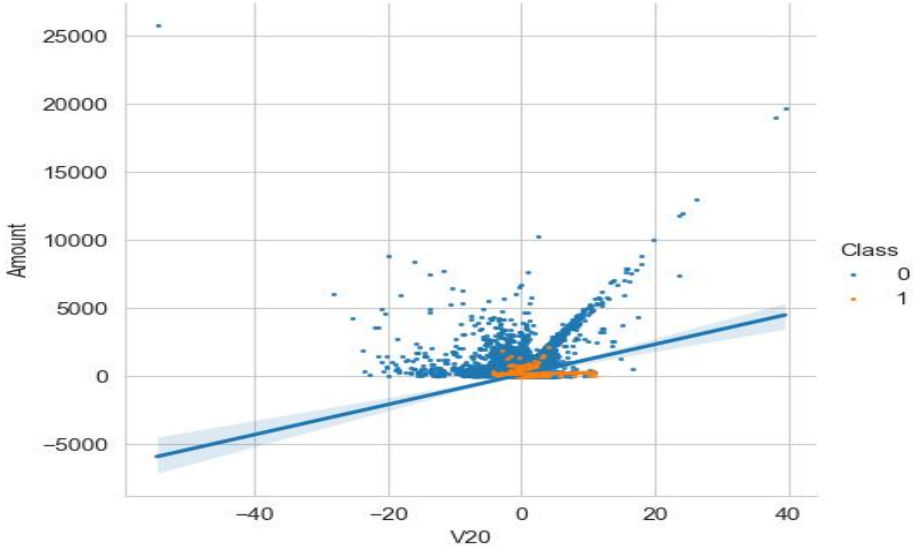
Batista, G. E., Prati, R. C., & Monard, M. C. (2004). A study of the behavior of several methods for balancing machine learning training data. *ACM SIGKDD Explorations Newsletter*, *6*(1), 20–29.

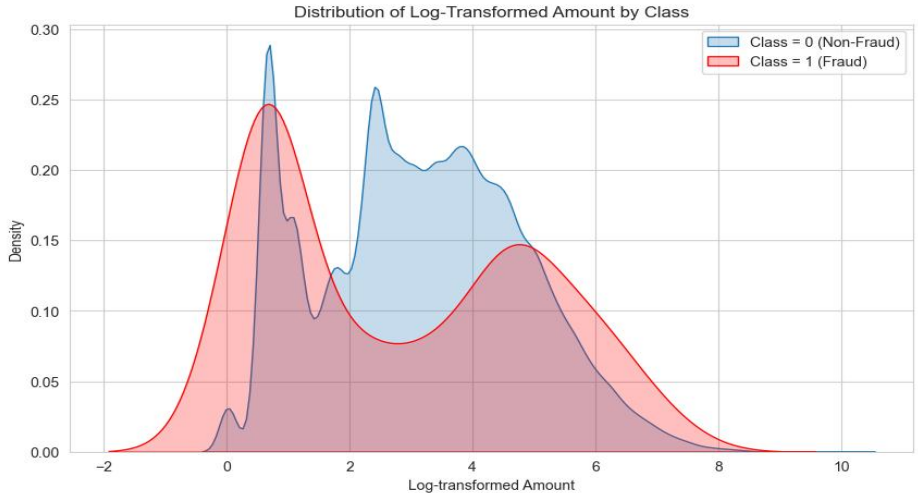
Molnar, C. (2022). *Interpretable machine learning: A guide for making black box models explainable*. (2nd ed.). https://christophm.github.io/interpretable-ml-book/

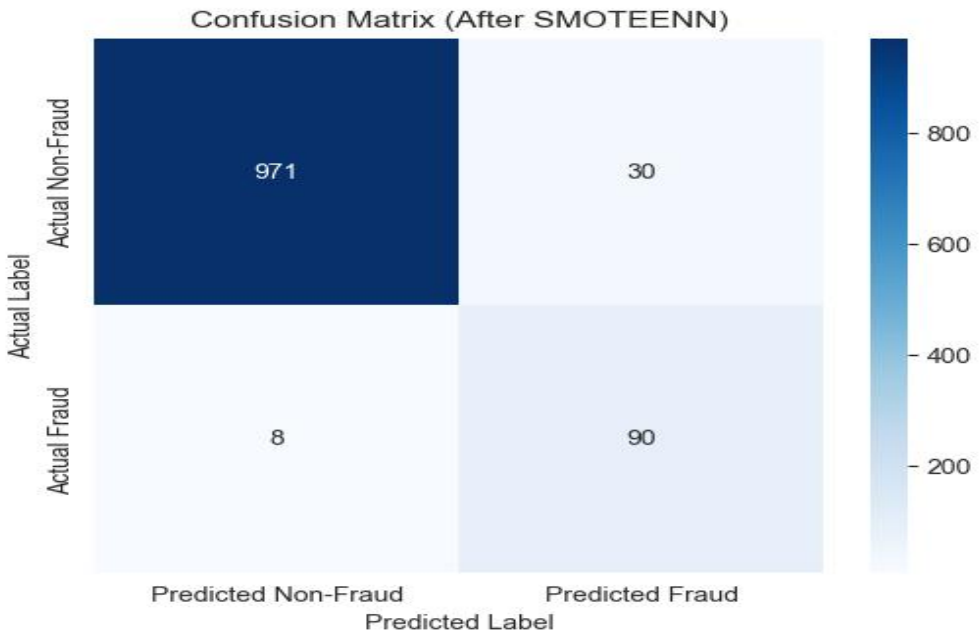
## 2 . Appendix

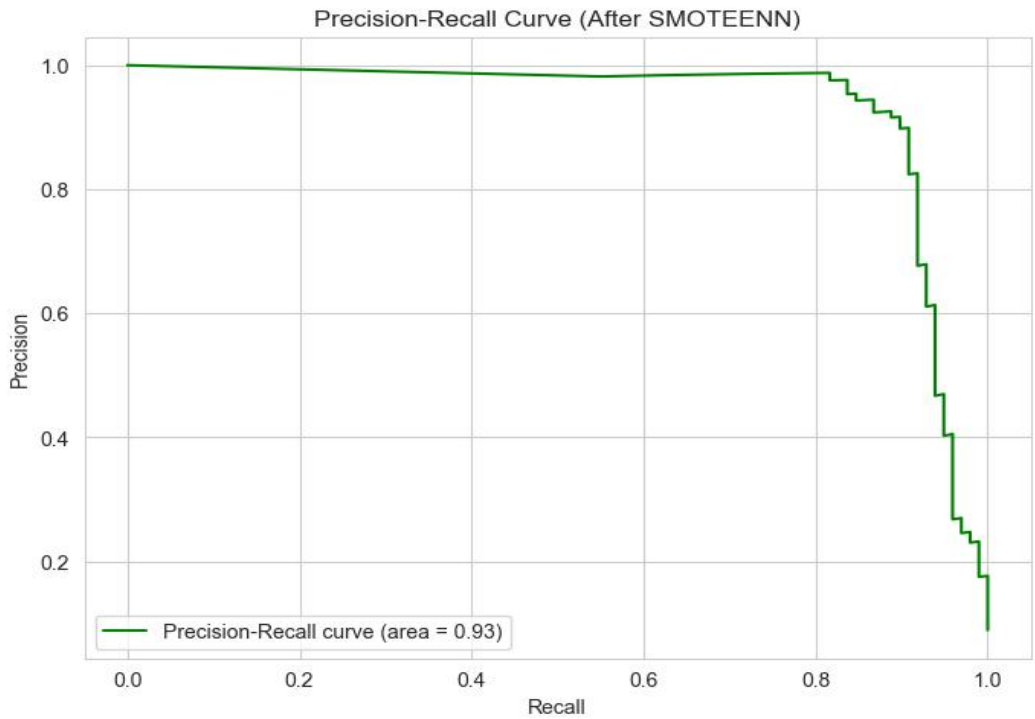

25000
20000
15000
10000
5000
0
Amount
−60
−40
−20
0
20
V2
Class
0
1


25000
20000
15000
10000
5000
0
Amount
−120
−100
−80
−60
−40
−20
0
20
40
V5
Class
0
1

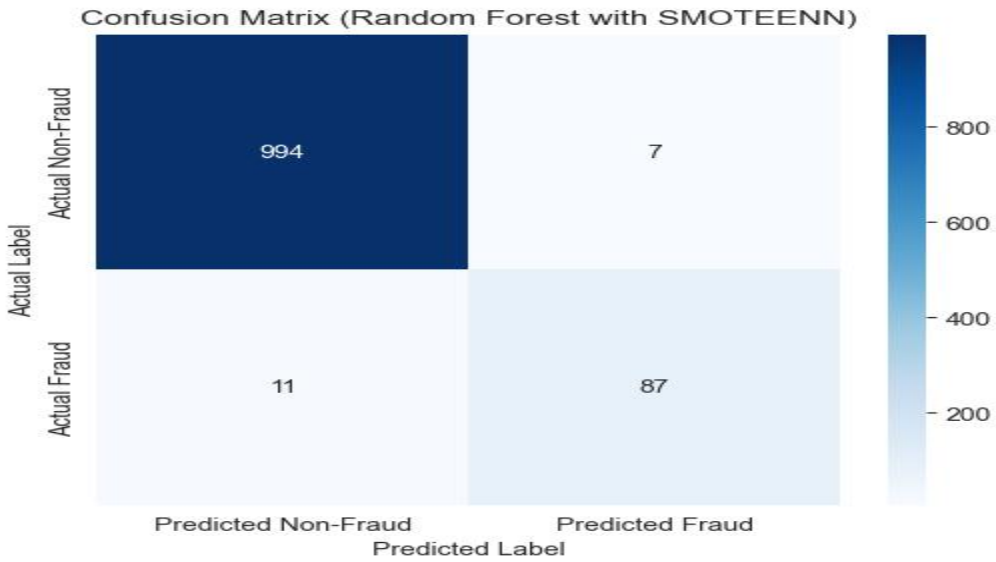

25000
20000
15000
10000
5000
0
−5000
Amount
−40
−20
0
20
40
V20
Class
0
1

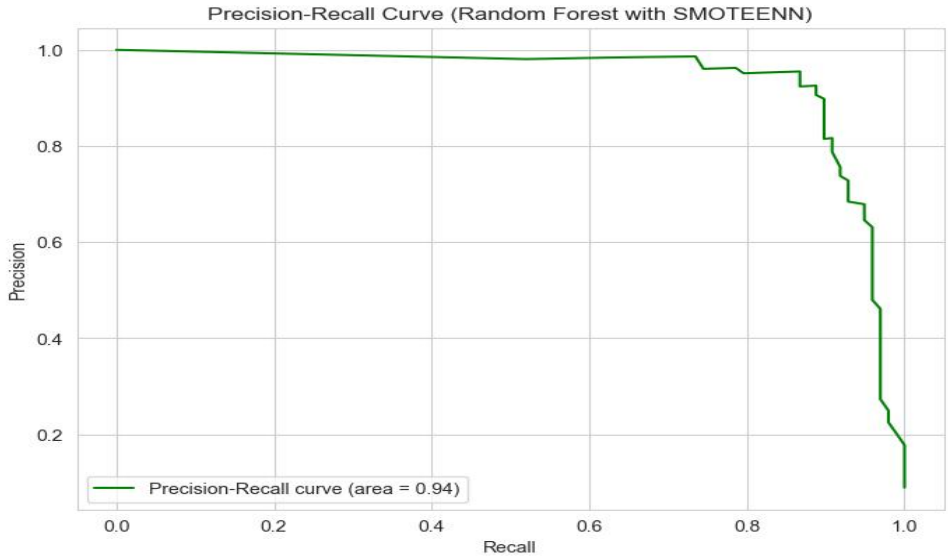

Distribution of Log-Transformed Amount by Class
Class = 0 (Non-Fraud)
Class = 1 (Fraud)
Density
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−2
0
2
4
6
8
10
Log-transformed Amount

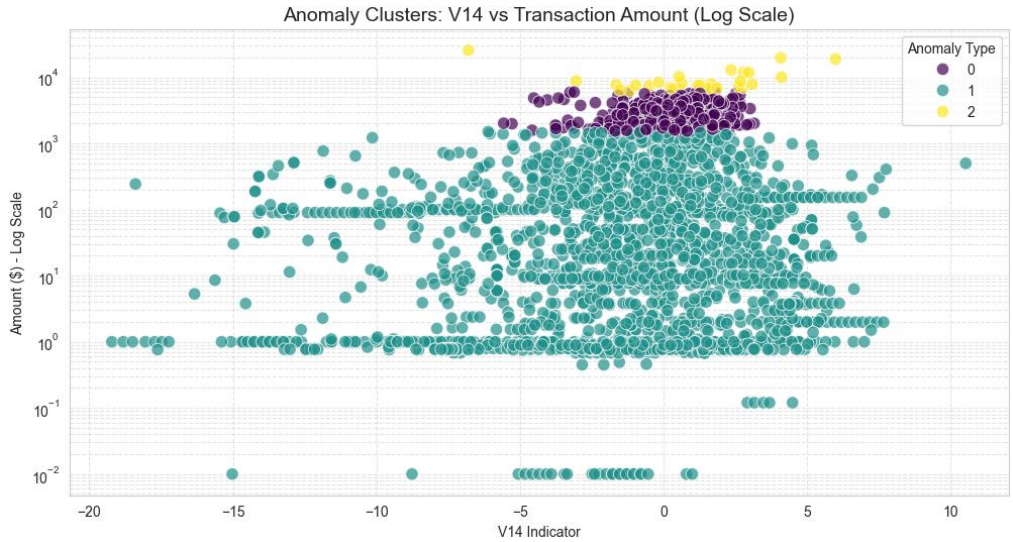

Confusion Matrix (After SMOTEENN)
Actual Non-Fraud
Actual Fraud
Actual Label
971
30
8
90
800
600
400
200
Predicted Non-Fraud
Predicted Fraud
Predicted Label

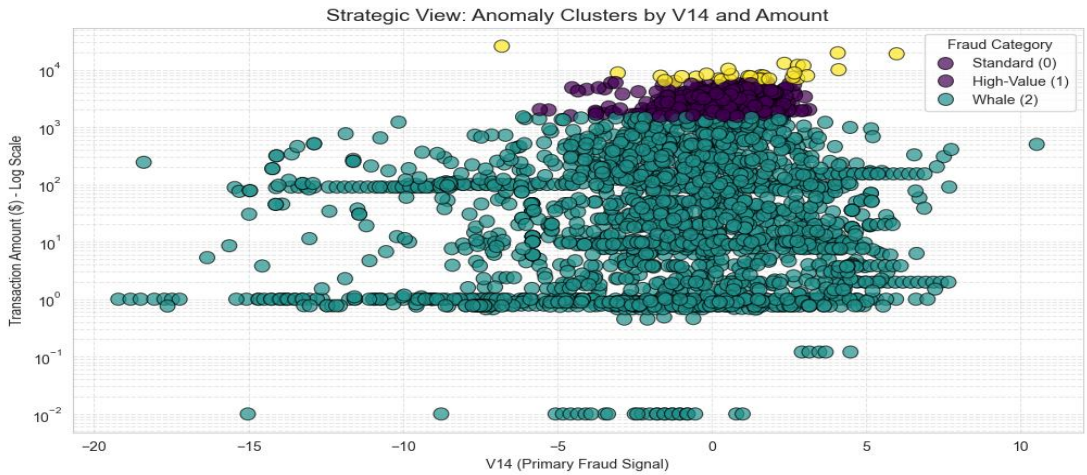


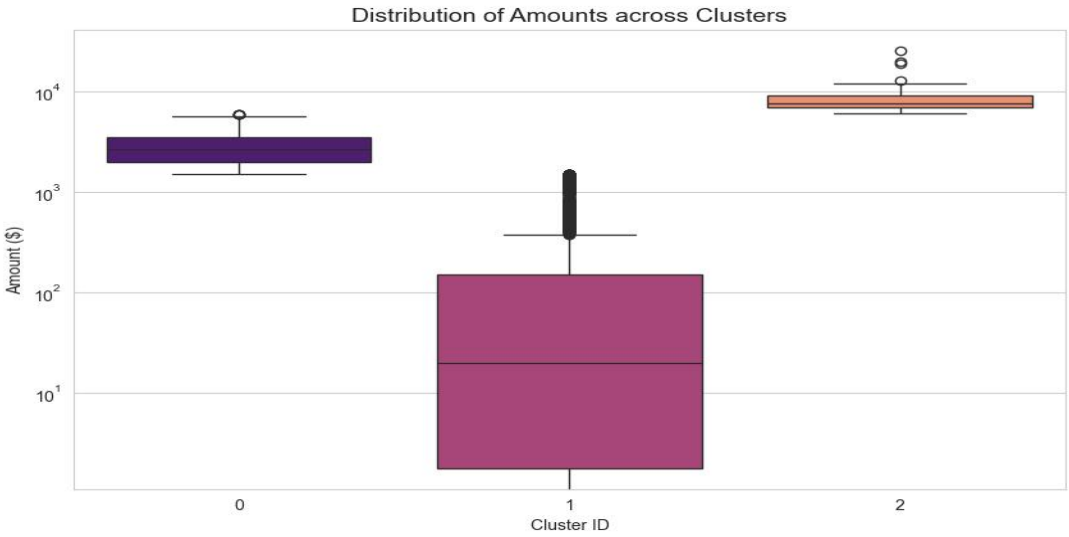

Confusion matrix and recall for logistic regression




Precision-Recall Curve (Random Forest with SMOTEENN)
Precision
Recall
Precision-Recall curve (area = 0.94)


Anomaly Clusters: V14 vs Transaction Amount (Log Scale)
Anomaly Type
0
1
2
Amount ($) - Log Scale
V14 Indicator


Strategic View: Anomaly Clusters by V14 and Amount
Fraud Category
Standard (0)
High-Value (1)
Whale (2)
Transaction Amount ($) - Log Scale
V14 (Primary Fraud Signal)



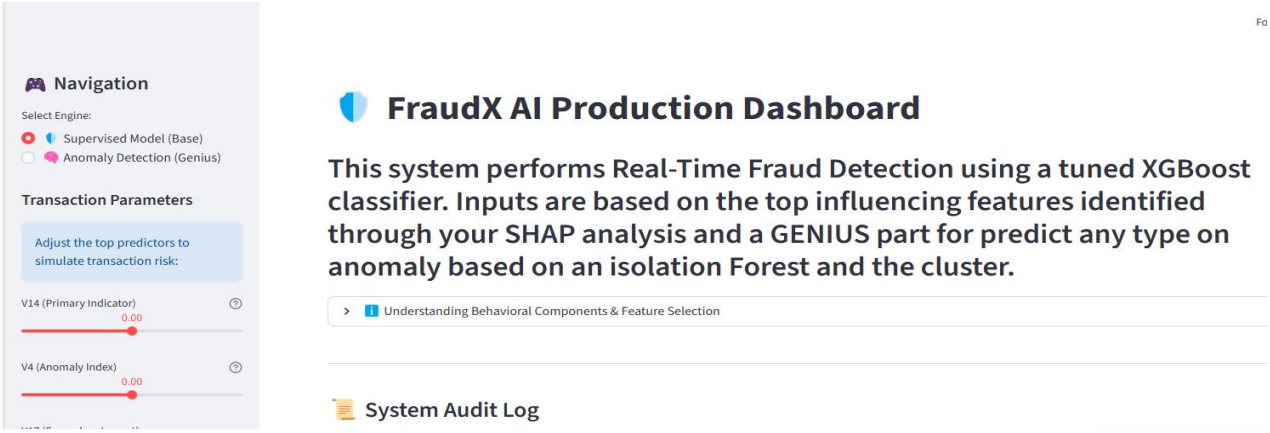

```
...
Classification Report (Random Forest with SMOTEENN):
              precision    recall  f1-score   support

           0       0.99      0.99      0.99      1001
           1       0.93      0.89      0.91        98

    accuracy                           0.98      1099
   macro avg       0.96      0.94      0.95      1099
weighted avg       0.98      0.98      0.98      1099

ROC AUC Score (Random Forest with SMOTEENN): 0.9848
```

```
Classification Report (After SMOTEENN):
              precision    recall  f1-score   support

           0       0.99      0.97      0.98      1001
           1       0.75      0.92      0.83        98

    accuracy                           0.97      1099
   macro avg       0.87      0.94      0.90      1099
weighted avg       0.97      0.97      0.97      1099

ROC AUC Score (After SMOTEENN): 0.9817
```

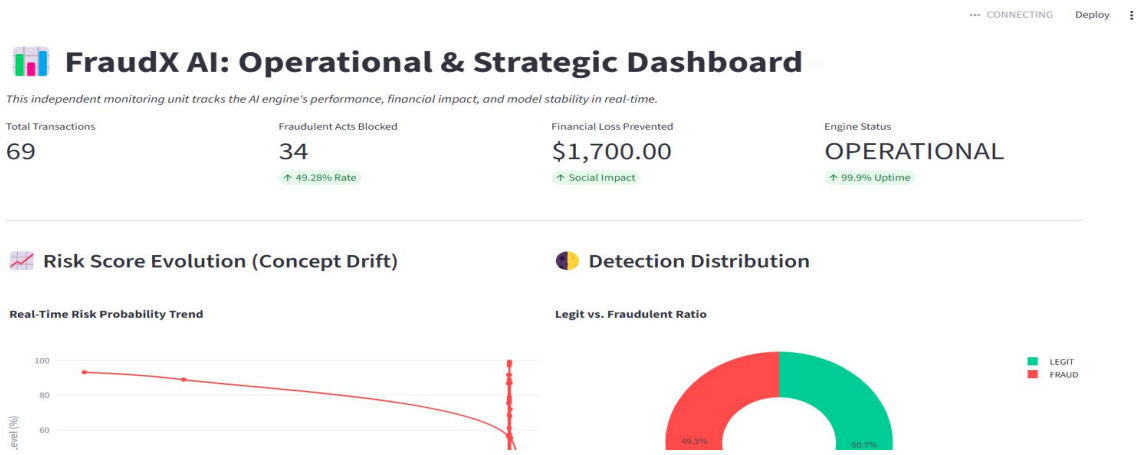
Streamlit_app and Github link to see the project

The applications  are available online (fraud/detection and monitoring) :

https://gramfraudanomalyanddetection-4qfvtrqibkebewpjrqrnyd.streamlit.app/




https://gramfraudmonitoring-clrpe2u2qt63afhfjynkmw.streamlit.app/




Github :

https://github.com/GRAMAYEBIE/Gram_Fraud_anomaly_and_detection